

Gayatri Nulaka

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Summary

Software Engineer and Researcher with robust data science and AI/ML expertise, experienced in building Large Language Models and Retrieval-Augmented Generation pipelines. Proven track record of optimizing systems and delivering scalable, data-driven solutions that enhance user experiences. Adept at collaborating with cross-functional teams to integrate AI solutions and drive measurable business impact.

EDUCATION

Master of Science in Computer Science **GPA: 3.9/4.0**

University of Texas at Arlington | Arlington, Texas

Course Work: Neural Networks ,Artificial intelligence ,Data Mining, ADV SE,ADV COA.

Bachelor of Technology(Data Science) in Computer Science **GPA: 3.3/4.0**

KL University | Vijayawada, India

Course Work: Data visualization, Data mining, SE, Artificial Intelligence, OS, Cloud computing.

PROFESSIONAL EXPERIENCE

Software Engineer-Walmart, Dallas, TX **Jan 2024 – Present**

- Engineered and optimized data integration microservices and RESTful APIs with Node.js, Python, and MongoDB, reducing data retrieval latency by 30% and scaling to 50K+ daily transactions across Walmart’s e-commerce platform.
- Integrated Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) pipelines (LangChain, OpenAI APIs) into enterprise support tools, enabling AI-driven knowledge retrieval and customer chatbots that reduced issue resolution time by 20%.
- Designed and deployed fine-tuned transformer models for product categorization and customer interaction data, improving recommendation accuracy and search relevance by 18%.
- Implemented CI/CD pipelines on AWS and Kubernetes, streamlining deployments, improving system reliability, and cutting release cycles from weeks to days.
- Built and executed end-to-end testing pipelines with Jest, Cypress, and React Testing Library, achieving 95% coverage and reducing production incidents by 40%.
- Partnered with cross-functional teams and product managers in Agile sprints to refine user stories, incorporate insights from 200+ support tickets, and deliver AI-driven features that boosted customer satisfaction by 15%.

Research Assistant-University of Texas at Arlington, Arlington, TX **Aug 2022 – Dec 2023**

- Designed and developed interactive research dashboards using React.js, D3.js, and TailwindCSS, enabling faculty and students to visualize large datasets and improving research data accessibility by 40%.
- Applied data mining and machine learning techniques (Python, Pandas, Scikit-learn) on research datasets, uncovering patterns and insights that supported peer-reviewed publications.
- Built and experimented with Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) pipelines (LangChain, OpenAI APIs), improving knowledge retrieval and research Q&A systems.
- Automated data preprocessing and ETL workflows, reducing manual cleaning effort by 60% and ensuring reproducibility of experiments for future researchers.
- Conducted exploratory data analysis (EDA) with Python and SQL, creating statistical models and visualizations that informed experimental design and hypothesis testing.
- Implemented unit and integration tests with Jest and Cypress to validate research prototypes, achieving 95% functional coverage and minimizing demo failures during lab presentations.
- Collaborated with professors and PhD candidates in Agile-style sprints, contributing to technical reports, data analysis, and prototype systems that were used in academic conferences.

- Built and optimized responsive web interfaces with **React.js, HTML5, CSS3, and JavaScript (ES6)**, improving page load performance by **30%** and enhancing usability for **5,000+ monthly users**.
- Developed reusable **React components** and implemented **state management** using Redux, reducing code duplication and cutting feature development time by **20%**.
- Integrated **RESTful APIs** on the frontend with Axios/Fetch, implementing efficient data fetching, caching, and error handling to deliver real-time product search and shopping features.
- Designed and executed **end-to-end tests with Jest and Cypress**, ensuring **95% test coverage** and reducing customer-reported UI bugs by **25%**.
- Partnered with product managers in **Agile sprints** to refine user stories, incorporating feedback from **200+ support tickets** into UI/UX updates, boosting customer satisfaction by **15%**.

PROJECTS

Student LMS

- Developed and optimized 15+ intricate UI components using React class components, enhancing user experience which led to a 40% reduction in load times and improved performance ratings across multiple applications.
- Optimized state management processes to enable real-time updates during user interactions, achieving a 30% improvement in application responsiveness and increasing user satisfaction scores by more than 25%.
- Engineered scalable RESTful APIs utilizing the Django REST Framework, resulting in a 40% reduction in data retrieval times and enhancing overall application performance for over 1,000 active users.

Leaf Disease Detection Using CNN

- Developed and deployed a Convolutional Neural Network (CNN) model utilizing TensorFlow and Keras, achieving a remarkable accuracy of 90%, which enhanced predictive performance by 35% compared to the previous model.
- Engineered a Flask-based web application that enables users to upload over 1,000 leaf images daily, resulting in a 90% increase in user engagement and providing real-time disease diagnostics with an accuracy rate of 95%.
- Streamlined model assessment techniques by integrating advanced metrics such as recall and F1-score, achieving an overall performance boost of 20% in machine learning applications while decreasing error rates from 8% to 5%.

Online Pizza Delivery System

- Developed a robust pizza ordering and delivery platform utilizing Java, React, and MongoDB, achieving a 40% reduction in order processing time while handling up to 1,500 simultaneous users.
- Engineered system performance enhancements that improved order update frequency by 75%, resulting in a 30% increase in customer satisfaction scores as measured through post-delivery surveys.
- Spearheaded the design and optimization of a mobile-first user interface, achieving a 40% increase in user engagement across mobile platforms while ensuring seamless cross-platform compatibility for over 10,000 users.

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, JavaScript, TypeScript, React, Node.js, Django REST Framework, Flask, SQL

Frontend Development: UI Development, Responsive Design, Component Libraries, Frontend Architecture, Wireframe to Production Implementation

Artificial Intelligence & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, OpenAI APIs, Machine Learning, Deep Learning, MLOps

Data Science & Data Mining: Data Mining, Exploratory Data Analysis (EDA), Feature Engineering, Scikit-learn, Pandas, NumPy, Statistical Modeling, SQL

Testing & Quality Assurance: Jest, Cypress, React Testing Library, Unit/Integration/End-to-End Testing, Code Review, Debugging & System Troubleshooting

Databases: MySQL, MongoDB, PostgreSQL, Oracle, DB2

DevOps & Cloud: AWS, Azure, GCP, Kubernetes, OpenShift, Cloud-Native Technologies, CI/CD Pipelines

Tools & Version Control: Git, GitHub, Jira